



Report

Corporate Bankruptcies of publicly traded firms in the US

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This Report file contains the explanations, while the Quarto (or HTML) document mainly includes code and R outputs.

1 Introduction

The goal of this project is to study corporate bankruptcies of publicly traded firms in the US and identify the most effective predictive models. The analysis proceeds in several steps. First, we perform data preparation and exploratory analysis using baseline models. Next, we incorporate additional firm-level variables inspired from the risk literature and macroeconomic indicators to expand our analysis. These new variables are evaluated through regression and traditional models. Finally, we apply machine learning techniques, including Lasso and Gradient Boosting, for variable selection and model optimization, determining the best-performing approach for bankruptcy prediction.

This study relies on multiple datasets:

- CRSP: Stock market data,
- Compustat: Financial and accounting data on publicly traded companies,
- Moody's annual default study: Data on default, losses and rating transitions for corporate bond, loan and deposit,
- Florida-UCLA-LoPucki Bankruptcy Research Database: Data on large public companies that have filed for Chapter 11 bankruptcy since October 1, 1979,
- FRED: Macro variables in the US.

By combining firm level financials, market data, default variables, and macroeconomic indicators, this project aims to evaluate the trade-off between model interpretability and predictive performance.

2 Data and methodology

2.1 Data preparation

In a technical point of view we use a parquet format for all of our databases, this allows us to manipulate easily huge databases. The first step of the data construction is to build a set of variables including the dependent variable Y , and explanatory variables X , from financial data of US companies. As the data are observed at various fiscal years end, this is called "firm year" observations.

The dependent variable is *default*. In the databases, there are two ways to detect the event of default, either if the company is in liquidation (Chapter 7) or in bankruptcy (Chapter 11). The *default* variable is binary:

$$default = \begin{cases} 1 & \text{if the firm is classified as Bankruptcy or Liquidation in any source,} \\ 0 & \text{otherwise.} \end{cases}$$

For explanatory variables, we first create a set of financial ratios on total assets, for instance Total liabilities on total assets, Working capital on total assets etc. All the variables are presented in Table 3 in the Appendix.

It has been decided to reduce the scope of the analysis: we keep the time period 1990-2023, as it covers different crisis (the Internet bubble, the 2008 financial crisis and the Covid-19 pandemic in 2020) and only three sectors of industry (among eleven) are selected:

- Retail

- Services
- Manufacturing

2.2 Outlier values

In case of too big impact of outliers, there are two different ways to deal with this issue : first we can use a log transformation of the problematic variables, or we can do a winsorization. The two ways are tried in the following sections. To determine if a variable needs to be winsorized, we look at the skewness and excess kurtosis. The skewness measures the asymmetry of a distribution. If it is close to 0, then the distribution is symmetric, if it is positive, there is a right tail, if it is negative, there is a left tail. A skewness between 0.5 and 1 reflects a moderate asymmetry. Skewness bigger than 1 reflects a strong asymmetry. The excess kurtosis quantifies the tailedness of a distribution compared to a normal distribution. It indicates the presence of outliers and the shape of the distribution's tails. If it is close to 0, tails are close to those of a normal distribution. If it is positive, there are fat tails (also called Leptokurtic Distribution), and if it is negative, there are thin tails (which is called Platykurtic Distribution).

2.3 General method

It has been decided to start with a simple model inspired from literature: the *Altman model*. In a second time, a revisited version of this model is considered. After, the Distance to default variable is added. Then, macro variables and market indicators are also considered in the models. Each time an improved version of the model is fitted to achieve, at the end, the more precise and complete model. The procedure is the same for each new model that are fitted for each of the three selected sectors:

0. New variables are constructed if needed
1. Descriptive statistics for each of the selected variables to identify necessity of winsorisation.
2. Variables for the model are selected and winsorized if necessary
3. Over-correlation is checked via correlation matrix between variables
4. Descriptive statistics for each of the selected (winsorized) variables and box plots to observe the individual features interaction with the response variable
5. Models are fitted
6. Model performances are checked via AUC and ROC curves

3 Altman Models

3.1 Original Altman models

Based on the procedure developed by Altman (1968) [1], we construct the Altman ratios (Z-scores), defined as follows :

$$\begin{aligned}
 X_1 &= \frac{\text{Working Capital}}{\text{Total Assets}}, \\
 X_2 &= \frac{\text{Retained Earnings}}{\text{Total Assets}}, \\
 X_3 &= \frac{\text{EBIT}}{\text{Total Assets}}, \\
 X_4 &= \frac{\text{Market Value of Equity}}{\text{Total Liabilities}}, \\
 X_5 &= \frac{\text{Sales}}{\text{Total Assets}}.
 \end{aligned}$$

The Altman ratios are relevant economic transformations as they are proxies for liquidity, solvency, profitability and operational efficiency.

We compute the first baseline logistic regression models for each of the three sectors using the five Altman ratios:

$$\begin{aligned} Default_k &= \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 X_{3k} + \beta_4 X_{4k} + \beta_5 X_{5k} + \epsilon_k, \\ k &\in \{\text{Manufacturing, Retail, Services}\} \end{aligned} \quad (1)$$

The results are presented in Table 4. Missing variables are not significant. We can observe that for the Retail sector 4 ratios out of 5 are significant. However, for the Manufacturing and Service sectors, the Altman ratios seem not to explain well the probability of default of a company. As the models are logistic regressions, we can only interpret the sign of the coefficients. Indeed, some results seem to be consistent: the more a company makes earnings, the less the probability to default. Similarly, a higher market value leads to a lower probability of default. The sign of working capital over total assets may seem unusual: here the more a company has working capital, the higher the probability of default. This could be explained by an inefficient working capital management for instance.

3.2 Altman revisited models 1

In a revisited version of the first procedure, Altman & Rijken (2004) [2] worked on a new version of the first model. This version includes two additional variables: the size and the age of the firm. Since there are multiple rows for each firm (one per year), these two variables change each year and impact differently the probability of default of the company. They are defined as :

$$\begin{aligned} \text{size} &= \frac{\text{Total liabilities}}{\text{Market value of Equity}}, \\ \text{age} &= \text{seniority of the firm in our sample.} \end{aligned}$$

Also, the model is now:

$$\begin{aligned} Default_k &= \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 X_{3k} + \beta_4 X_{4k} + \beta_5 \text{size}_k + \beta_6 \text{age}_k + \epsilon_k, \\ k &\in \{\text{Manufacturing, Retail, Services}\} \end{aligned} \quad (2)$$

3.3 Altman revisited models 2

In order to deal with outliers, Altman & Rijken (2004) [2] worked on log transformations of the initial ratios. They define :

$$\begin{aligned} x_i &= -\ln(1 - X_i), \text{ for } i = 2, 3 \\ x_4 &= 1 + \ln(X_4). \end{aligned}$$

And

$$\begin{aligned} Default_k &= \beta_1 X_{1k} + \beta_2 x_{2k} + \beta_3 x_{3k} + \beta_4 x_{4k} + \beta_5 \text{size}_k + \beta_6 \text{age}_k + \epsilon_k, \\ k &\in \{\text{Manufacturing, Retail, Services}\} \end{aligned} \quad (3)$$

3.4 Altman revisited models 1 & 2 winsorized

To reduce further the risk of bias introduced by outliers, in a last step, two last version of these models are considered using winsorization. Then the models 2 and 3 are refitted using only winsorized variables. Results are respectively illustrated in Table 5, Table 6 and Table 7, Table 8 for models with winsorization.

3.5 Detailed Interpretation – Altman revisited model 2 winsorized : Retail Sector Example

In this section, the entire procedure is detailed for the Retail sector for the *Altman revisited model 2 winsorized*.

0. **New variables are constructed:** In this section, we build the log transformation of Altman ratios.

1. **To fit this new model, we select the log transformed variables, and winsorize them:** Indeed, the descriptive statistics in Table 9 reveal significant skewness and extreme values in several key financial variables, indicating the presence of outliers. For instance, size and age shows high maximum values relative to their medians, with skewness and kurtosis far from one, reflecting a heavy-tailed distribution. Similarly, firm market value display extreme observations that can disproportionately influence mean-based measures and regression results. To mitigate the impact of these extreme observations and improve the robustness of statistical analyses, it is therefore necessary to apply winsorization, reducing the influence of outliers while preserving the overall structure of the dataset.
2. **Over-correlation is checked** via the correlation matrix Figure 7 between variables, in this case, we don't find highly correlated variables. So no need to delete variables.
3. **The descriptive statistics** for the Retail sector in Table 10 show more reasonable variation among firms thanks to the winsorization. Most firms in the retail sector are relatively young (median = 12 years) and exhibit a low default risk. Logarithmic transformations of financial variables mitigate the impact of extreme values. Overall, the sector is dominated by a few large firms, while the majority remain modest in size.
4. **The results of the model** are in the second column of Table 8 The Altman revisited model for the Retail sector highlights several significant financial determinants of bankruptcy risk. Working capital over total assets has a positive and highly significant coefficient (3.105, $p < 0.01$), indicating that firms with higher liquidity are more likely to default. Log-transformed EBIT has a negative coefficient ($-3.600, p < 0.01$), suggesting that higher operational profitability decreases bankruptcy risk in Retail sector. Log market value is also negative and significant ($-0.344, p < 0.01$), meaning larger firms in terms of market capitalization are less prone to default. The constant term ($-3.298, p < 0.01$) sets the baseline risk level, while other variables such as age and $\ln_{re_at}$ were not significant in this sector.

Overall, the model is estimated on 5,215 Retail firms, with a log-likelihood of -650.212 and an AIC of $1,308.424$, indicating a reasonable fit. The results suggest that liquidity, operational profitability, and firm size are the primary drivers of bankruptcy risk for firms in the Retail sector.
5. **Model performance is evaluated using the Area Under the Curve (AUC) and ROC curves.** For the Retail sector, the Altman revisited model achieves an AUC of 0.696, indicating a moderate ability to discriminate between firms that will default and those that will not. The ROC curve further illustrates the trade-off between true positive and false positive rates, confirming that while the model captures important risk signals (liquidity, profitability, and firm size), its predictive power is limited compared to more advanced machine learning models (see later on the Lasso and Gradient Boosting performances).

4 Models with additional variables

4.1 Models including distance to default

To flesh out this project, another idea is to create a *Distance to default* (DtD) variable. Based on Bharath & Shumway (2008) [3], DTD is a model based measure for gauging how far a firm is away from default. Indeed, the DtD is a predictor based on the market value of assets and the company's debt. Therefore, this variable incorporates market capitalization and stock price volatility, capturing real time default risk. For simplicity, it is decided to use a "Naïve" method for DtD rather than an iterative one. Then, DtD is constructed following a Merton-type structural model by creating components of the variable and then applying the formula to obtain the "Naïve" DtD.

The components are:

$$\begin{aligned}
E &= \text{Market Equity} \\
DP &= \text{Default Point using Duffie definition} \\
\sigma_v &= \text{Total volatility} \\
r_{i,t-1} &= \text{expected return of asset} \\
T &= \text{Default horizon, here 1 year}
\end{aligned}$$

Using the formula :

$$DtD^{\text{Naïve}} = \frac{\ln\left(\frac{E+DP}{DP}\right) + (r_{i,t-1} - 0.5 \sigma_v^2) T}{\sigma_v \sqrt{T}}.$$

For each of the firm we obtain daily values, we decide to aggregate these values to obtain one unique DtD per year, per firm. We winsorize the DtD variable at 1-99 % to lower the impact of outliers. As before, we compute the two Altman revisited winsorized models, adding the DtD variable.

The models are:

$$\begin{aligned}
Default_k &= \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 X_{3k} + \beta_4 X_{4k} + \beta_5 size_k + \beta_6 age_k + \beta_7 DtD_k + \epsilon_k, \\
k &\in \{\text{Manufacturing, Retail, Services}\}
\end{aligned} \tag{4}$$

and,

$$\begin{aligned}
Default_k &= \beta_1 X_{1k} + \beta_2 x_{2k} + \beta_3 x_{3k} + \beta_4 x_{4k} + \beta_5 size_k + \beta_6 age_k + \beta_7 DtD_k + \epsilon_k, \\
k &\in \{\text{Manufacturing, Retail, Services}\}
\end{aligned} \tag{5}$$

Table 11 and Table 12 illustrates the results. Most variables are highly significant. Moreover, the Distance to Default variable is also significant at the 1 % level and negative for the three sectors. This means that the furthest a firm is from default, the lower its probability of default.

4.2 Models including macro and market variables

We also add macroeconomic and market variables to capture the exogenous impact of the business cycle on firms. A firm may be solvent but any major macroeconomic shock could lead a company to bankruptcy. Table 13 summarizes all the added variables. Macroeconomic variables are obtained from the Federal Reserve Economic Data (FRED) database and aggregated at the annual level using yearly averages. Book equity is constructed from Compustat following the Fama-French methodology and set to missing when non-positive.

After obtaining these new variables, one macro (GDP) and one market (BE) variables are added to the previous models. The equations are:

$$\begin{aligned}
Default_k &= \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 X_{3k} + \beta_4 X_{4k} + \beta_5 size_k \\
&\quad + \beta_6 age_k + \beta_7 DtD_k + \beta_8 GDP + \beta_9 BE_k + \epsilon_k, \\
k &\in \{\text{Manufacturing, Retail, Services}\}
\end{aligned} \tag{6}$$

and,

$$\begin{aligned}
Default_k &= \beta_1 X_{1k} + \beta_2 x_{2k} + \beta_3 x_{3k} + \beta_4 x_{4k} + \beta_5 size_k \\
&\quad + \beta_6 age_k + \beta_7 DtD_k + \beta_8 GDP + \beta_9 BE_k + \epsilon_k, \\
k &\in \{\text{Manufacturing, Retail, Services}\}
\end{aligned} \tag{7}$$

Table 14 illustrates the results obtained for each of the three sectors. Most variables are significant. The two additional variables (GDP and BE) are both mostly significant at the 1 % level and negative. The BE is not significant in the retail sector for the first version of the model. The GDP is not significant in the Services sector for the second version of the model (with logs), otherwise, it is significant which means that the better the macroeconomic conditions, the lower the probability to default.

5 Machine Learning Models

Following WRDS industry financial ratio, more than 70 financial ratios are constructed. They are classified into seven main groups:

- Capitalization
- Efficiency
- Financial Soundness/Solvency
- Liquidity
- Profitability
- Valuation
- Others

5.1 Lasso Models

To find the best models, we consider these 70 ratios (Altman ratios are included in the 70), the macro variables and DtD. We winsorize all the financial ratios and remove the variables which are highly correlated. We need to select the most adapted variables to fit our models. We employ a two-step estimation procedure.

First, variables are selected using LASSO with the penalty parameter chosen by cross-validation. Then, a Lasso with a k-fold cross-validation procedure is conducted to determine the optimal λ by sector. Indeed, choosing the right λ is crucial as it determines the penalization level. The higher it is, the more it will shrink the coefficients to 0. We set the number of folds in the Cross-Validation to 5. We obtain the following values for lambdas :

Sector	λ_{min}	λ_{1se}
Manufacturing	0.00007	0.00251
Retail	0.00062	0.00841
Services	0.00065	0.00731

Table 1: Optimal λ - Lasso

λ_{min} minimizes the cross-validation error. The model is therefore more complex, with more variables selected. λ_{1se} is the highest λ for which the CV error is within one standard deviation of the minimum error. The model is therefore simpler, but more robust.

Table 16 summarizes the variables selected by the model using the λ_{min} for each sector.

Second, coefficients are re-estimated using a ridge-penalized logistic regression on the selected set of variables. This approach mitigates issues related to multicollinearity and quasi-complete separation while retaining the variable selection properties of LASSO.

Table 17 reports the coefficients estimated by the post-LASSO Ridge regression for each sector. Given the large number of variables, we do not discuss each coefficient individually. Nonetheless, the reported values provide an overview of the relative importance of the variables selected by the LASSO and stabilized by Ridge regression.

Overall, the LASSO procedure efficiently selected a subset of relevant predictors for each sector, reducing dimensionality while keeping the most informative variables for the next Ridge regression and model evaluation.

5.2 Gradient Boosting Models

Now, we want to predict the probability of default by using an ensemble classifier: we use Gradient Boosting. It builds a predictive model as a sum of weak learners estimated sequentially. At each iteration, a new tree is fitted to the residual errors of the previous model by minimizing the gradient of a specified loss function. A learning rate controls the contribution of each tree, helping to prevent overfitting. As it is based on the combination of several trees, we decide of the optimal number of boosting iterations with the Out-of-Bag improvement criterion. For the manufacturing sector, we use then 1135 trees, 1178 trees for the Retail sector and 645 for Services. As for the other hyperparameters, we selected:

distribution	interaction.depth	n.minobsinnode	shrinkage	bag.fraction
Bernoulli	2	10	0.01	0.7

Table 2: Gradient Boosting Hyperparameters

First, all the required variables are gathered into a single dataset, including the variables used in the initial models as well as the macroeconomic and market variables. A first Gradient Boosting model is then trained, and the optimal number of trees is determined using an out-of-sample performance criterion. Based on this optimal number of trees, predicted probabilities and performance metrics are computed. In parallel, an optimal pruning level is selected in order to derive an interpretable tree-based representation of the Gradient Boosting model, allowing for the selection of the final model specification. The resulting trees of this procedure are Figure 8, Figure 9 and Figure 10.

In manufacturing, pe_op_basic ¹ is the strongest risk indicator. Firms with $pe_op_basic < 6.25$ have a high event probability (30.27%), while those above this threshold show a much lower risk (0.26–6.27%). Additional factors such as $liabs_assets$ ², be , and advertising-to-sales ratio influence risk, particularly for recent years. High leverage combined with low operational performance signals the greatest vulnerability (18.01%).

In retail, pay_turn ³ drives risk. Companies with $pay_turn < -1.62$ face a high probability of adverse events (11.8%), with risk levels varying depending on inventory intensity (inv_act) and turnover measures. Conversely, firms with strong interest coverage ($intcov_ratio \geq 2.23$) and efficient stock management have lower risk (0.76–4.01%).

For the services sector, ROA ⁴ is the primary risk factor. Firms with $roa < -0.0775$ exhibit high risk (14%), particularly if highly leveraged ($debt_at \geq 0.77$). Those with positive ROA and moderate debt show low risk (0.2–7.9%). Operational profitability (pe_op_basic) and liquidity ratios ($cash_debt$) further refine the assessment.

Across sectors, risk is the highest for poorly performing and highly leveraged firms. Manufacturing risk is tied to operational performance and leverage, retail risk to accounts payable and inventory management, and services risk to profitability and debt levels. Despite sector-specific drivers, high leverage and weak financial performance are universal predictors of vulnerability.

6 Assessing classification power

Finally, ROC curves and AUC are used in order to compare all the models in two ways: first by making a comparison of the same model between the three sectors, second by making a comparison of different models in a same sector. Between sectors the following graphic shows the comparison for the Altman models.

¹ pe_op_basic : Price-to-Operating Earnings Per Share (EPS), excluding exceptional items, on a basic basis.

² $liabs_assets$: total liabilities on total assets

³ pay_turn : accounts payable turnover

⁴ ROA : Return on Assets

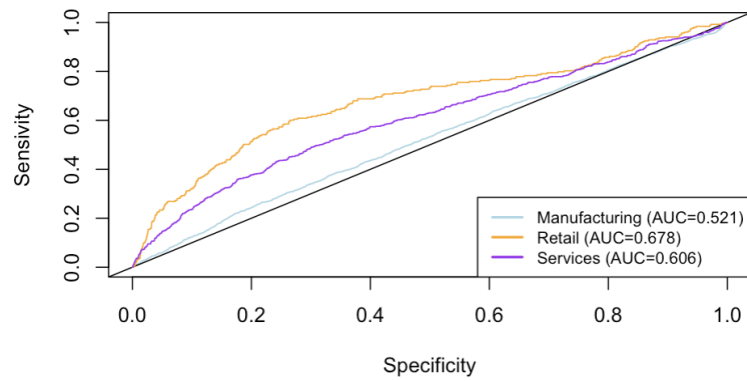


Figure 1: ROC Curves and AUC - Altman baseline model

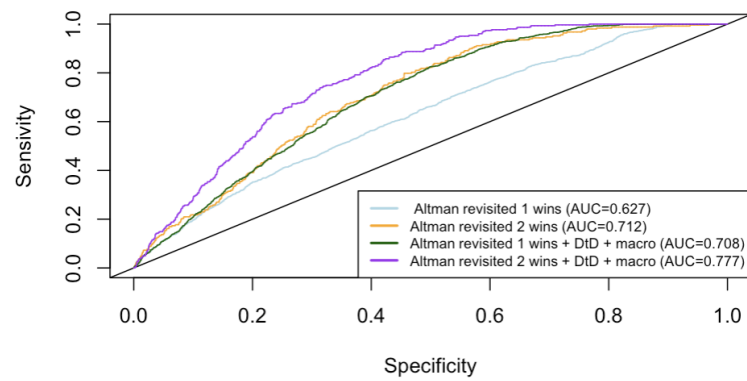


Figure 2: ROC Curves and AUC - Manufacturing sector

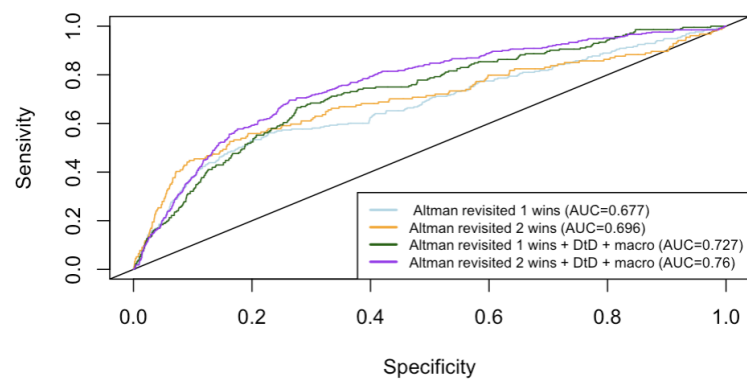


Figure 3: ROC Curves and AUC - Retail sector

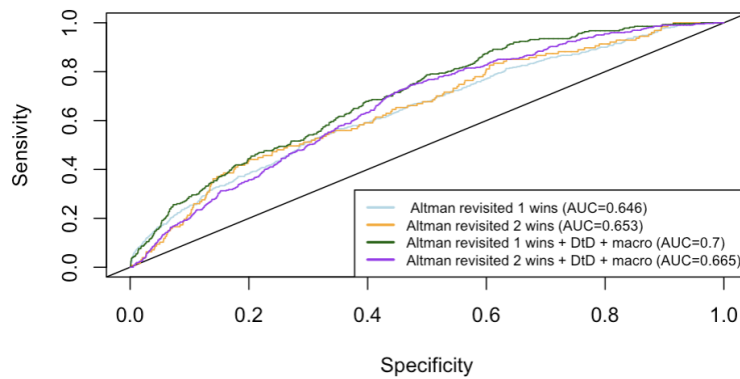


Figure 4: ROC Curves and AUC - Service sector

For the Altman baseline models, the overall predictive performance is modest, as reflected by relatively low AUC values. Despite this, the Retail sector stands out, showing the best performance among the three sectors. Adding the size and age variables increases the classification power (models with log transformation and winsorization) but remains low. Adding the Distance to Default and macroeconomic variables increases the AUC, which reaches 0.777 for the Manufacturing sector. However, it remains low for the Service sector: 0.7.

We observe that the ROC Curves and AUCs are the highest for the Gradient Boosting models but not necessarily for the Lasso models :

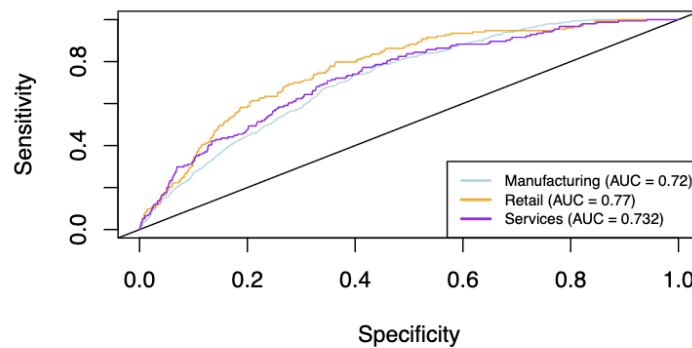


Figure 5: ROC Curves and AUCs - Lasso

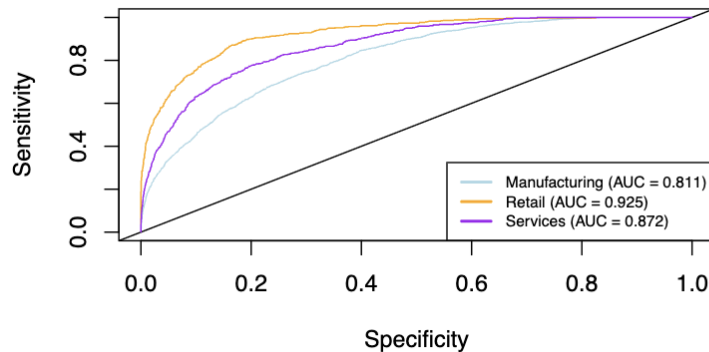


Figure 6: ROC Curves and AUCs - Gradient Boosting

Machine learning models significantly improve credit risk prediction compared to traditional Altman baselines. While Altman models show modest AUCs, Retail performs best. Lasso raises AUCs to 0.770 (Retail), 0.732 (Services), and 0.720 (Manufacturing). Gradient Boosting achieves the highest performance, with AUCs of 0.925 (Retail), 0.872 (Services), and 0.811 (Manufacturing), making it a highly effective classifier. These results demonstrate that advanced models capture sector-specific risk patterns much better than traditional financial ratio approaches.

7 Conclusion

To conclude this project, Table 18 summarizes the results obtained across the different models. Gradient Boosting achieves the highest AUC for all three sectors. Lasso provides the second-best performance for the Retail and Services sectors, while the Altman revisited + DtD + Macro model performs best for Manufacturing.

Overall, these findings highlight a clear trade-off between interpretability and predictive accuracy. Gradient Boosting delivers superior classification performance but relies on 75 features, making individual interpretation challenging. Simpler and more structured models, such as the different versions of the Altman model, remain valuable due to their ease of interpretation.

Finally, for predicting bankruptcy of public firms in the US, we strongly recommend using Gradient Boosting. It would also be worthwhile to explore other machine learning methods and compare their performance against Gradient Boosting.

8 References

References

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9 Appendix

9.1 Altman models

Table 3: Definition of Financial Variables

Variable	Description	Definition
<i>wcap_at</i>	Working capital to assets	WC/AT
<i>seq_at</i>	Equity to assets	SEQ/AT
<i>lt_at</i>	Total liabilities to assets	LT/AT
<i>lct_at</i>	Current liabilities to assets	LCT/AT
<i>che_at</i>	Cash holdings to assets	CHE/AT
<i>rect_at</i>	Receivables to assets	$RECT/AT$
<i>inv_t_at</i>	Inventories to assets	$INVT/AT$
<i>act_at</i>	Current assets to assets	ACT/AT
<i>ppent_at</i>	PPE to assets	$PPENT/AT$
<i>intan_at</i>	Intangible assets to assets	$INTAN/AT$
<i>sale_at</i>	Sales to assets	$SALE/AT$
<i>cogs_at</i>	Cost of goods sold to assets	$COGS/AT$
<i>xsga_at</i>	SG&A expenses to assets	$XSGA/AT$
<i>dp_at</i>	Depreciation to assets	DP/AT
<i>ebit_at</i>	EBIT to assets	$EBIT/AT$
<i>xint_at</i>	Interest expenses to assets	$XINT/AT$
<i>ni_at</i>	Net income to assets	NI/AT
<i>re_at</i>	Retained earnings to assets	RE/AT
<i>capx_at</i>	Capital expenditures to assets	$CAPX/AT$
<i>dv_at</i>	Dividends to assets	DV/AT
<i>ebitda</i>	Earnings before interest, taxes, depreciation and amortization	$EBIT + DP$
<i>ebitda_at</i>	EBITDA to assets	$EBITDA/AT$
<i>mktval_at</i>	Market value to assets	$MKVALT/AT$
<i>mktval_lt</i>	Market value to liabilities	$MKVALT/LT$

Table 4: Altman model for each sectors

Var.	(Manufacturing)	(Retail)	(Services)
wcap_at	0.034**(0.016)	1.721*** (0.318)	-0.001(0.003)
re_at	-0.004**(0.002)	—	—
ebit_at	—	-0.614** (0.239)	—
mktval_lt	—	-0.108*** (0.026)	—
sale_at	—	0.287*** (0.056)	—
Const.	-3.511*** (0.025)	-3.808*** (0.165)	-3.800*** (0.041)
Obs.	65,777	6,454	27,464
LogLik	-8,745.242	-1,028.421	-2,891.388
AIC	17,496.480	2,066.842	5,786.776

*p<0.1; **p<0.05; ***p<0.01

Table 5: Altman revisited model

Var.	(Manufacturing)	(Retail)	(Services)
wcap_at	—	1.698*** (0.321)	—
ebit_at	—	-0.798*** (0.233)	—
mktval_lt	—	-0.102*** (0.026)	—
size	—	—	—
age	-0.016*** (0.002)	0.018*** (0.004)	-0.036*** (0.005)
Const.	-3.234*** (0.035)	-3.568*** (0.144)	-3.429*** (0.062)
Obs.	66,470	6,419	28,491
LogLik	-8,837.122	-1,030.857	-2,952.129
AIC	17,678.240	2,071.715	5,908.257

*p<0.1; **p<0.05; ***p<0.01

Table 6: Altman model revisited with logs

Var.	(Manufacturing)	(Retail)	(Services)
wcap_at	1.206*** (0.380)	2.004*** (0.315)	—
ln_re_at	-0.692*** (0.224)	—	-1.314*** (0.419)
ln_ebit_at	1.850*** (0.528)	—	—
ln_mktval_lt	-0.550*** (0.066)	-0.449*** (0.043)	—
age	-0.016*** (0.004)	—	-0.034*** (0.010)
size	—	—	—
Const.	-3.328*** (0.166)	-3.081*** (0.092)	-3.809*** (0.150)
Obs.	20,172	6,458	12,431
LogLik	-1,270.848	-1,010.219	-690.453
AIC	2,553.696	2,026.437	1,386.905

*p<0.1; **p<0.05; ***p<0.01

Table 7: Altman model revisited with winsorization

Var.	(Manufacturing)	(Retail)	(Services)
wcap_at	—	1.794*** (0.320)	—
ebit_at	-0.199*** (0.043)	-0.743*** (0.243)	-0.223*** (0.073)
mktval_lt	—	-0.075*** (0.024)	—
size	0.056*** (0.004)	0.044*** (0.008)	0.058*** (0.005)
age	-0.015*** (0.002)	0.019*** (0.004)	-0.030*** (0.006)
Const.	-3.502*** (0.051)	-3.802*** (0.153)	-3.723*** (0.085)
Obs.	45,645	6,419	20,117
LogLik	-5,319.769	-1,018.491	-1,860.737
AIC	10,647.540	2,048.983	3,729.474

*p<0.1; **p<0.05; ***p<0.01

Table 8: Altman model revisited with logs and winsorization

Var.	(Manufacturing)	(Retail)	(Services)
wcap_at	1.247*** (0.388)	3.105*** (0.425)	—
ln_re_at	-0.880*** (0.240)	—	-1.333*** (0.419)
ln_ebit_at	2.082** (0.870)	-3.600*** (1.394)	—
ln_mktval_lt	-0.543*** (0.069)	-0.344*** (0.080)	—
age	-0.015*** (0.005)	—	-0.034*** (0.010)
Const.	-3.337*** (0.176)	-3.298*** (0.166)	-3.805*** (0.150)
Obs.	20,172	5,215	12,431
LogLik	-1,272.145	-650.212	-690.595
AIC	2,556.290	1,308.424	1,387.190

*p<0.1; **p<0.05; ***p<0.01

9.2 Detailed Interpretation – Altman revisited models 2 winsorized : Retail Sector Example

Table 9: Descriptive statistics – Retail sector

Variable	n	Min	P5	P25	Median	Mean	P75	P95	Max	SD	Skew	Kurt
default	9963	0.000	0.000	0.000	0.000	0.057	0.000	0.000	1.000	0.232	3.821	12.598
wcap_at	9817	-5.176	-0.434	0.017	0.171	0.177	0.338	0.560	0.984	0.252	-2.583	40.807
ln_re_at	6493	0.000	0.006	0.139	0.306	0.438	0.551	1.295	9.392	0.504	4.210	35.222
ln_ebit_at	7898	0.000	0.004	0.062	0.103	0.122	0.157	0.286	1.716	0.097	3.207	23.804
ln_mktval_lt	6512	-6.782	-2.273	0.612	1.516	1.424	2.314	3.460	8.097	1.355	-0.493	1.563
size	6511	0.001	0.032	0.269	0.597	2.985	1.474	6.546	3561.961	34.558	53.725	2396.179
age	9927	0.000	1.000	6.000	12.000	16.240	24.000	44.000	64.000	13.275	1.135	0.650

Table 10: Descriptive statistics – Retail sector variables windsorized

Variable	n	Min	P5	P25	Median	Mean	P75	P95	Max	SD	Skew	Kurt
default	9963	0.000	0.000	0.000	0.000	0.057	0.000	0.000	1.000	0.232	3.821	12.598
wcap_at	9817	-5.176	-0.434	0.017	0.171	0.177	0.338	0.560	0.907	0.252	-2.594	40.898
ln_re_at	6493	0.004	0.006	0.139	0.306	0.414	0.551	1.295	1.672	0.383	1.597	2.370
ln_ebit_at	7898	0.003	0.004	0.062	0.103	0.121	0.157	0.286	0.533	0.087	1.694	4.307
ln_mktval_lt	6512	-2.820	-2.273	0.612	1.516	1.431	2.314	3.460	6.559	1.326	-0.308	0.555
size	6511	0.004	0.032	0.269	0.597	1.819	1.474	6.546	45.003	4.736	6.717	52.423
age	9927	0.000	1.000	6.000	12.000	16.154	24.000	44.000	52.000	13.017	1.042	0.241

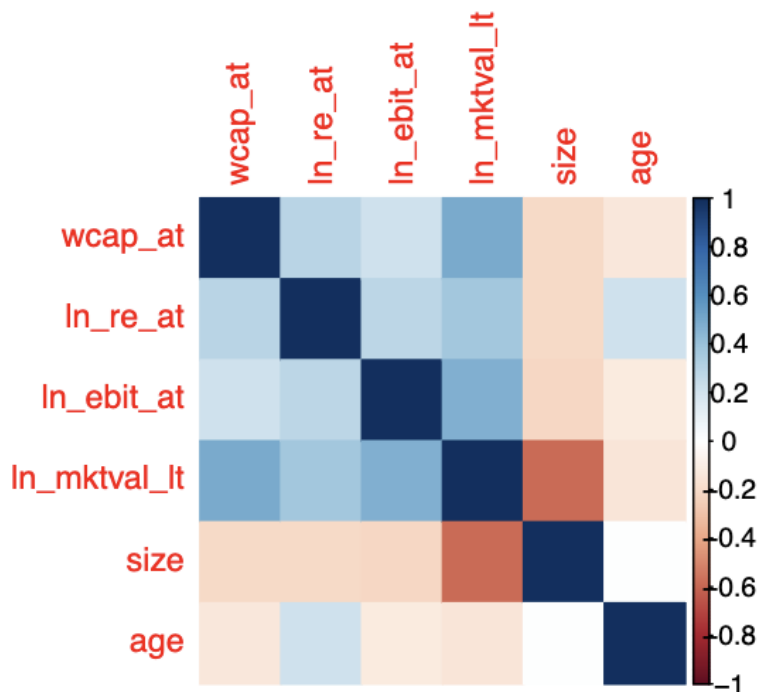


Figure 7: Correlation Matrix

9.3 Models with additional variables

Table 11: Altman revisited model including Naive DtD

Var.	(Manufacturing)	(Retail)	(Services)
wcap_at	—	1.550*** (0.324)	—
re_at	-0.008* (0.004)	—	-0.019*** (0.006)
mktval_lt	0.003*** (0.002)	—	—
size	0.041*** (0.005)	0.039*** (0.005)	0.049*** (0.007)
age	-0.006** (0.002)	0.029*** (0.005)	-0.022*** (0.006)
DtD	-0.223*** (0.016)	-0.295*** (0.038)	-0.181*** (0.028)
Const.	-2.811*** (0.070)	-3.132*** (0.179)	-3.182*** (0.119)
Obs.	42,165	6,124	18,049
LogLik	-4,746.857	-932.660	-1,597.565
AIC	9,505.714	1,875.320	3,205.131

*p<0.1; **p<0.05; ***p<0.01

Table 12: Altman revisited with logs model including Naive DtD

Var.	(Manufacturing)	(Retail)	(Services)
wcap_at	—	2.888*** (0.436)	—
ln_re_at	-0.564*** (0.212)	—	—
ln_mktval_lt	-0.227*** (0.058)	—	-0.179*** (0.041)
ln_ebit_at	—	-4.005*** (1.438)	—
size	—	0.031* (0.017)	—
age	-0.010** (0.004)	-0.021*** (0.006)	-0.026*** (0.007)
DtD	-0.213*** (0.032)	-0.218*** (0.050)	-0.175*** (0.030)
Const.	-2.467*** (0.139)	-3.316*** (0.262)	-2.714*** (0.116)
Obs.	21,943	5,047	18,086
LogLik	-1,499.787	-605.542	-1,615.813
AIC	3,009.573	1,223.083	3,239.626

* p<0.1; ** p<0.05; *** p<0.01

Table 13: Macroeconomic and Market Variables

Variable	Level	Definition
<i>t10y</i>	Macro (annual)	Average 10-year U.S. Treasury yield
<i>t3m</i>	Macro (annual)	Average 3-month U.S. Treasury yield
<i>yield_curve</i>	Macro (annual)	Yield curve slope: $t10y - t3m$
<i>GDP</i>	Macro (annual)	Real U.S. Gross Domestic Product
<i>unemployment</i>	Macro (annual)	U.S. unemployment rate
<i>cpi</i>	Macro (annual)	Consumer Price Index (CPI)
<i>rate_ff</i>	Macro (annual)	Federal Funds effective rate
<i>baa_yield</i>	Macro (annual)	Moody's Baa corporate bond yield
<i>be</i>	Firm-year	Book equity (Fama–French definition)

Table 14: Altman model including Naive DtD, GDP and Book Equity

Var.	(Manufacturing)	(Retail)	(Services)
re_at	—	—	-0.038*** (0.008)
ebit_at	-0.293*** (0.075)	—	—
wcap_at	—	1.497*** (0.361)	-0.751*** (0.251)
mktval_lt	—	-0.069** (0.032)	—
size	0.049*** (0.008)	—	0.056*** (0.010)
age	0.005* (0.003)	0.029*** (0.006)	0.016** (0.008)
DtD	-0.118*** (0.017)	-0.216*** (0.044)	-0.101*** (0.032)
GDP	-0.0001*** (0.0001)	-0.0001*** (0.0003)	-0.0001*** (0.0003)
BE	-0.001*** (0.0001)	—	-0.0003*** (0.0001)
Constant	-1.612*** (0.215)	-0.735 (0.491)	-2.096*** (0.408)
Obs.	39,920	5,762	16,302
LogLik	-4,098.863	-840.154	-1,326.115
AIC	8,211.727	1,694.307	2,668.229

* p<0.1; ** p<0.05; *** p<0.01

Table 15: Altman model including Naive DtD, GDP and Book Equity (log)

Var.	(Manufacturing)	(Retail)	(Services)
ln_re_at	−0.750*** (0.208)		
ln_ebit_at	−0.289*** (0.066)		
wcap_at	1.645*** (0.284)	1.645*** (0.284)	
ln_mktval_lt		−6.861*** (1.140)	
age	0.026*** (0.005)	0.026*** (0.005)	
DtD	−0.088*** (0.033)	−0.062** (0.029)	−0.186*** (0.026)
GDP	−0.0001*** (0.00003)	−0.0002*** (0.00002)	
BE	−0.001*** (0.0001)		−0.001** (0.0001)
Constant	−0.790* (0.420)	0.073 (0.306)	−3.184*** (0.087)
Obs.	21,882	7,405	23,286
LogLik	−1,446.819	−1,216.135	−2,038.198
AIC	2,905.637	2,444.271	4,082.397

*p<0.1; **p<0.05; ***p<0.01

Table 16: Variables selected by LASSO at $\lambda_{\min}$ by sector

Variable	Manufacturing	Retail	Services
pstk_new	✓	✓	
be	✓	✓	
bm	✓	✓	✓
ch_act	✓	✓	✓
ch_che	✓	✓	✓
ch_lct	✓		
ch_dlc	✓	✓	✓
ch_txp	✓	✓	✓
ocf	✓	✓	✓
evm	✓	✓	✓
ps	✓	✓	
pcf	✓	✓	✓
dpr	✓	✓	✓
npm	✓	✓	
gpm	✓	✓	✓
ptpm	✓	✓	
roa	✓	✓	✓
roe	✓	✓	✓
roce	✓	✓	
efftax	✓	✓	✓
aftret_eq	✓	✓	✓
aftret_invcapx	✓		✓
pretret_noa	✓	✓	
pretret_earnat	✓	✓	✓
gprof	✓	✓	✓
equity_invcap	✓		✓
totdebt_invcap	✓	✓	✓
int_totdebt	✓	✓	✓
cash_lt	✓		✓
inv_t_act	✓	✓	✓
rect_act	✓	✓	✓
debt_at	✓		✓
debt_ebitda	✓	✓	✓
short_debt	✓	✓	✓
curr_debt	✓	✓	
lt_debt	✓	✓	✓
profit_lct	✓	✓	
ocf_lct	✓	✓	
cash_debt	✓	✓	✓
fcf_ocf	✓	✓	✓
dltt_be	✓	✓	✓
liabs_assets	✓	✓	✓
debt_capital	✓	✓	✓
de_ratio	✓	✓	✓
cash_ratio	✓	✓	
cash_conversion	✓		
at_turn	✓	✓	✓
rect_turn	✓	✓	
pay_turn	✓	✓	✓
sale_invcap	✓	✓	✓
sale_equity	✓	✓	✓
sale_nwc	✓		✓
rd_sale	✓	✓	✓
adv_sale	✓	✓	✓
staff_sale	✓	✓	
avg_at	✓	✓	
accruals	✓	✓	✓
capei	✓	✓	✓
year	✓	✓	✓
t10y	✓	✓	
t3m	✓	✓	
unemployment	✓	✓	✓
baa_yield	✓		
DtD	✓	✓	✓

Table 17: Post-LASSO Ridge coefficients by sector

Variable	Manufacturing	Retail	Services
pstk_new	0.0888	0.1808	
be	-0.5084	0.2192	
bm	0.0594	0.1243	0.0228
ch_che	-0.0896	-0.1467	-0.0403
ch_lct	-0.1193		
ch_dlc	-0.0336	0.0734	-0.0366
ch_txp	0.0753	0.0839	-0.1754
ocf	-0.4738	-0.4218	-0.3465
evm	-0.0226	-0.1356	-0.1411
ps	-0.0543		
pcf	0.0341	-0.0605	0.0862
dpr	-0.1002	-0.1344	-0.0578
npm	-0.1144	0.0716	
gpm	0.0584	-0.1388	0.1135
ptpm	0.0215	-0.0814	
roa	-0.1970	-0.2626	-0.2025
roe	-0.0011	-0.1420	-0.1628
roce	-0.0182	0.0537	
efftax	-0.2904	-0.1653	-0.2638
aftret_eq	-0.0667	0.0250	0.0742
aftret_invcapx	-0.1149		0.0557
pretret_noa	0.0410	-0.0825	
pretret_earnat	0.0887	-0.0802	0.0884
gprof	-0.1266	0.1373	0.0420
equity_invcap	-0.0039		0.0923
totdebt_invcap	0.0156	-0.1601	0.0932
int_totdebt	-0.0119	0.0282	0.0395
cash_lt	0.1604		0.0701
inv_t_act	0.0567	0.3826	0.0304
rect_act	-0.0570	-0.0662	0.1200
debt_at	0.0449		0.4486
debt_ebitda	0.0559	0.1847	0.1513
short_debt	0.0412	-0.0466	0.1178
curr_debt	-0.2099	-0.4993	
lt_debt	-0.1146	-0.2789	-0.0823
profit_lct	-0.0980	-0.0831	
ocf_lct	-0.0154	-0.0643	
cash_debt	0.0394	0.1744	-0.1429
fcf_ocf	0.0724	0.0312	0.1291
dltt_be	-0.0092	0.2098	0.1788
liabs_assets	0.0584	-0.1565	-0.1365
debt_capital	0.0866	0.1845	0.0438
de_ratio	0.0352	-0.0677	-0.0682
cash_ratio	-0.0309	-0.1236	
cash_conversion	-0.0711		
at_turn	0.0377	0.0269	0.0404
rect_turn	0.0300	-0.0605	
pay_turn	-0.0894	-0.1666	0.0488
sale_invcap	0.0088	0.1559	0.0241
sale_equity	-0.0409	-0.1401	-0.0416
sale_nwc	0.0452		-0.0369
rd_sale	-0.1073	-0.0981	0.0790
adv_sale	0.0474	0.0514	0.0717
staff_sale	-0.0702	-0.0767	
avg_at	-0.1148	0.0712	
accruals	-0.0702	-0.2698	
capei	0.0139	-0.0459	0.0848
year	-0.4584	-0.2853	-0.2272
t10y	-0.0108	0.1410	
t3m	-0.0727	0.0756	
unemployment	0.0258	0.1454	0.1647
baa_yield	-0.0576		
DtD	-0.2083	-0.1548	-0.2974
ch_act		-0.0898	-0.0592

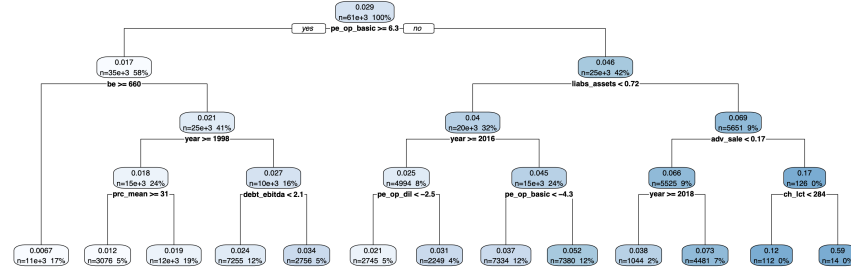


Figure 8: Tree using Gradient Boosting for Manufacturing sector

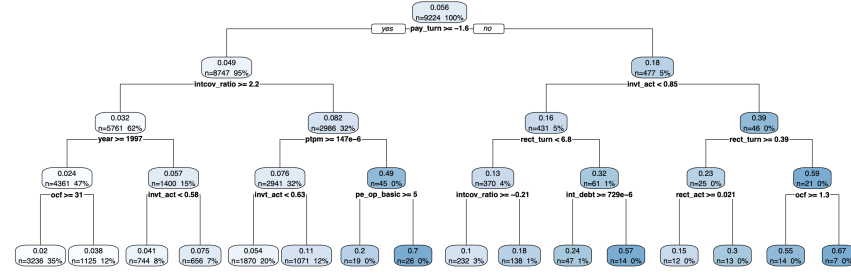


Figure 9: Tree using Gradient Boosting for Retail sector

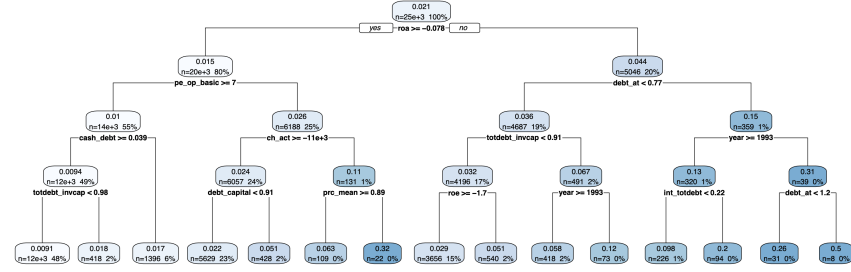


Figure 10: Tree using Gradient Boosting for Services sector

9.4 Models comparison

Table 18: Comparison of AUC across models and sectors

Model	Manufacturing	Retail	Services
Altman revisited	0.627	0.677	0.646
Altman revisited with logs	0.712	0.696	0.653
Altman revisited + DtD + Macro	0.708	0.727	0.700
Altman revisited with logs + DtD + Macro	0.777	0.760	0.665
Lasso	0.720	0.770	0.732
Gradient Boosting	0.811	0.925	0.872

Notes: All the models are winsorized